MILITARY SPECIFICATION

COATING, INTERIOR, FUEL RESISTANT, FOR GASOLINE CONTAINERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an organic protective coating to be used on the interior of steel fuel containers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-N-95	-	Naphtha; Aliphatic
VV-B-231	-	Benzene (Benzol), Technical
PPP-P-1892	-	Paint, Varnish, Lacquer and Related Materials Packaging, Packing and Marking of

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MIL-G-3056	-	Gasoline, Automotive, Combat

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MIL-G-5572 - Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-T-5624 - Turbine Fuel Aviation, Grades JP-4 and JP-5

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials

D 381 - Standard Method of Test for Existent Gum in Fuels by Jet Evaporation

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

The Clean Air Act

Title 42 United States Code Sections 1857 et. seq.
and regulations promulgated thereunder
by the Administrator of the Environmental Protection Agency

(Copies of the Act may be obtained from the Environmental Protection Agency, Regional Office, John F. Kennedy Building, Boston, MA 02203.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Material. The interior coating shall contain titanium dioxide pigmented epoxy or epoxy phenolic type resin requiring baking to achieve cure. The manufacturer is given wide latitude in the selection of the other component materials to produce a coating formulation which meets the requirements of this specification. The manufacturer shall comply with the "Clean Air Act" in the selection of his components (see 2.2).

3.2 Effect on fuels in storage. The stored fuels shall show no increase in gum content when tested as specified in 4.2.2.

3.3 Durability of coating. When tested as specified in 4.2.2, the coating shall not soften, blister, wrinkle, peel, lose adhesion or show rusted areas totaling more than two square inches or other evidence of failure.

3.4 Effect on gum content in aircraft gasoline. The test fuel shall show no increase in gum content as a result of contact with the baked coating when tested as specified in 4.2.2.

3.5 Impact resistance. When tested as specified in 4.2.2, the coating shall not crack or peel.

3.6 Flexibility. The coating shall not crack, either at room temperature or at low temperature, when tested as specified in 4.2.2.

3.7 Resistance to boiling water. The coating shall show no blistering, softening, lifting, loss of adhesion, cracking, whitening, evidence of rusting of the panel or other differences in appearance from the coating on the non-immersed panel, when tested as specified in 4.2.2.

3.8 Inertness. The total gain or loss in weight shall not exceed 8 milligrams when tested as specified in 4.2.2.

3.9 Color. The cured coating shall be any light color. The directional reflectance shall be not less than 30 percent when tested as specified in 4.2.2.

3.10 Opacity. The coating shall be opaque when tested as specified in 4.2.2.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certification of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, materials and components shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Testing of the end item. The coating shall be tested for the characteristics listed in table I. A certificate of compliance will be acceptable for identification of compound, effect on fuels and durability of coating to the extent specified in table I. The sample for all tests shall be one quart composite, made up of equal amounts selected at random throughout the lot in frequency listed below:

Lot size (gallons)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Failure of the coating in any test characteristic shall be cause for rejection of the lot. All test reports shall contain the individual values utilized in expressing the final result.

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TABLE I Instructions for testing the end item

Characteristic	Specification reference		Requirements applicable to		Number determinations composite sample	Results reported as	
	Requirement	Test method	Individual unit	Composite sample		Pass or fail	Numerically to nearest
Identification of compound	3.1	1/	-	-	-	-	-
Effect on fuels in storage (each type fuel specified)	3.2	4.3.1 1/	-	X	1	X	-
Durability of coating	3.3	4.3.1 1/	-	X	1	X	-
Effect on gum content in aircraft gasoline	3.4	4.3.2.6	-	-	-	X	-
Impact resistance	3.5	4.3.2.2	-	X	1	X	-
Flexibility at 65° to 80°F.	3.6	4.3.2.3	-	X	1	X	-
Flexibility at -20°F.	3.6	4.3.2.3	-	X	1	X	-
Resistance to boiling water	3.7	4.3.2.4	-	X	1	X	-
Inertness	3.8	4.3.2.5	-	X	2	-	milligram
Reflectance	3.9	4.3.2.7	-	X	1	-	percent
Opacity	3.10	4.3.2.8	-	X	1	X	-

1/ The supplier shall furnish a certificate of compliance with the first lot and with each lot in which there is a change in the formulation, stating that the coating meets the requirements in 3.1, 3.2, and 3.3 The certificate shall be accompanied by actual test, inspection or other verifiable data. For each lot subsequent to the test, in which there is no change in the formulation, the supplier shall submit a certificate of compliance to this effect.

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* 4.2.3 Inspection of preparation for delivery. Packaging, packing, marking and palletization (as applicable) shall be inspected in accordance with the quality assurance provisions in PPP-P-1892.

4.3 Test methods.

4.3.1 Effect on fuels in storage and durability of the coating.

* 4.3.1.1 Specimens. Four 5-gallon, low carbon steel cans shall be used for testing. The cans shall be commercial, closed top, screw cap, twenty-four gauge steel, 5-gallon containers. Interior surfaces shall have been treated in accordance with type I or type II surface preparation of TT-C-490. The interior surfaces shall be free from oil. The coating shall have been applied to the interior surfaces and cured prior to the double seaming of the bottom. The coating shall cover the entire interior surface of the can including the exterior surface of the vent tube and the inner face of the plug. Care shall be exercised that the countersink of the can bottom and the corresponding area of the can body are thoroughly coated. The vent tube interior need not be coated. The dry film thickness shall be 0.6 mil. plus or minus 0.1 mil except that up to 1.2 mils. thickness will be permitted in limited areas which are difficult to coat uniformly, such as lap seams, vent tube and embossings.

4.3.1.2 Fuels. The following three fuels shall be used: jet engine fuel conforming to MIL-T-5624, grade JP-4; aircraft engine fuel conforming to MIL-G-5572, grade 100/130 and combat automotive gasoline conforming to MIL-G-3056, type I. Just prior to the storage test, each fuel shall be tested to confirm that its gum content is within specification limits. The gum content shall be the average of two determinations.

* 4.3.1.3 Procedure. For each of the three fuels specified in 4.3.1.2, a specimen container shall be filled to its rated capacity and sealed with its gasket and tightened closure. The containers shall be stored for one month at a temperature of 60° to 90°F. At the conclusion of the storage period, the stored fuels in the specimen containers shall be sampled and tested (2 determinations per sample) for gum content in accordance with ASTM D 381. For each fuel, the average of the gum content determination at the start shall be deducted from the average of the gum content determinations at the end of the storage period. The value thus obtained shall be compared with the appropriate reproducibility value in "Precision" table of ASTM D 381. If the calculated value is equal to or less than the amounts permitted in the table, the test result shall be reported as "no increase in gum content". The containers shall then be emptied and the coating examined for evidence of rusting, softening, blistering, wrinkling, peeling, loss of adhesion or other coating failure.

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4.3.2 Tests using laboratory prepared panels.

4.3.2.1 Preparation of panels. Twenty gauge cold rolled steel panels shall be surface cleaned and phosphate coated in accordance with TT-C-490, type I or type II. Then the coating under test shall be applied, dried and cured in accordance with the coating manufacturer's directions to give a dry film thickness of 0.6 ± 0.1 mil., except the one 3 by 6-inch panel for the opacity test shall have a dry film thickness of 1.2 mils. ± 0.1 mil. Panels for the inertness and effect on gum content tests (4.3.2.5 and 4.3.2.6) shall be coated on both sides and edges whereas panels for other tests need be coated on one side only. All coated panels shall be free of fatty edges, sags or other application defects. Coated and cured panels for all tests shall be aged for 24 hours at 73.5° ± 4°F and 50 ± 5 percent relative humidity, before testing.

4.3.2.2 Impact resistance. An approximately 3 by 6-inch test panel shall be impacted by means of a two-pound, one inch diameter round nose, steel rod dropped from a height of 14 inches. A guide tube shall be used in aligning the drop weight and measuring the dropping distance. The diameter of the rounded impacting end of the steel rod shall be one half inch (see 6.3). The coated side of the panel shall be face down over a hollow area and the impact shall be delivered to the side panel opposite the test coating. The impacted area shall be examined for evidence of cracking or peeling.

4.3.2.3 Flexibility. An approximately 3 by 6-inch coated panel shall be bent slowly (3 to 4 seconds) at room temperature (65° to 80°F.) through 180° over 1/4-inch diameter mandrel with the coated side outward. The coating shall then be examined for evidence of cracking. Another similar panel with a 1/2-inch diameter mandrel shall be conditioned for 4 hours at minus 20° ± 2°F. and then tested at that temperature in the same manner over the 1/2-inch diameter mandrel. This coating shall also be examined for evidence or cracking.

4.3.2.4 Resistance to boiling water. A coated panel, approximately 3 by 6 inches shall be completely immersed in boiling distilled water for 2 hours. After cooling to room temperature, the panel shall be immediately compared with a nonimmersed panel for compliance with the requirements specified in 3.7

4.3.2.5 Inertness. Three weighed 1 by 4-inch, coated panels shall be positioned vertically in a 500 ml. Erlenmeyer flask fitted with a ground glass connected Friedrichs internal reflux condenser. The panels shall be arranged so as not to be in flat contact with each other. A 70/30 by volume mixture of naptha conforming to TT-N-95, type I and benzene conforming to VV-B-231, grade 1 shall be poured into the flask to the level of the center of the panel length. The panels shall be boiled in the solvent mixture for 7 hours and then removed, dried at 200°F.. for 1 hour in an oven, cooled and reweighed. Weight shall be determined to the nearest milligram. The total gain or loss in weight of the three panels combined shall be determined.

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* 4.3.2.6 Effect on gum content in aircraft gasoline. Two 1 by 5-inch coated panels shall be immersed without face to face contact in an amber colored 29-mm. by 300-mm. test tube containing 65 ml. of aircraft engine gasoline conforming to MIL-G-5572, grade 100/130. The test tube shall be equipped with an internal reflux condenser which has a double bend mecury seal at the top to prevent escape of vapors. The test tube shall be immersed 3 inches in a water bath maintained at 140° to 150°F. The test shall consist of 5 alternate cycles of 7 hours in the water bath and 17 hours of exposure to room temperature (65° to 80°F.). After the 5-day cycle, the gum determination shall be made in accordance with ASTM D 381. A blank determination (gasoline without coated panels) shall be run simultaneously and deducted from the value obtained for the panel exposure. If the difference is within the repeatability tolerance allowed in the precision table of ASTM D 381, the test result shall be reported as "no increase" in gum content.

4.3.2.7 Reflectance. The directional reflectance shall be determined in accordance with Method 6121 of FED-STD-141.

4.3.2.8 Opacity. A 3 by 6-inch panel prepared as specified in 4.3.2.1 and having a dry film thickness of 0.6 ± 0.1 mil. shall be visually compared for opacity with a 3 by 6-inch panel prepared as specified in 4.3.2.1 and having a dry film thickness of 1.2 ± 0.1 mil. Opacity shall be determined in accordance with the comparison of color of Method 4250 of FED-STD-141.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A or C. Coating, in the quantity specified (see 6.2), shall be packaged in accordance with the applicable requirements of PPP-P-1892.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A, B, or C. Coating, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-P-1892.

* 5.3 Marking. Marking shall be in accordance with the military requirements of PPP-P-1892.

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5.4 Precautionary marking. Container of the coating shall display the following precautionary marking:

CAUTION

May be irritating to skin, eyes and respiratory tract.
Avoid prolonged or repeated breathing of vapor.
Use only with adequate ventilation.
Avoid contact with skin and eyes.
If eye contact occurs, flush immediately with large ammounts of water.
Keep container closed when not in use.

6. NOTES

6.1 Intended use. The coating covered by this specification is intended for use on the 5-gallon military gasoline can and 55-gallon military gasoline drums.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Quantity required (see 5.1.1).

6.3 Impact tester. A variable impact tester, 28-inch-pound range of the type referenced in 4.3.2.2, is manufactured by Gardner Laboratory Inc., Bethesda, MD 20014.

6.4 Marginal notations. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 8010-0780

Review activities:

Army - AT, MD, MR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52032C(ME)	Coating, Interior, Fuel Resistant, for Gasoline Containers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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